A Reproducible Four-Qubit Quantum-Fuzzy Logical Petri Net: Operator Construction, Independent Verification, and Ordering Sensitivity

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Abstract

Reproducibility of hybrid quantum–classical models is limited not only by numerical precision but also by conventions that are frequently left implicit. This paper develops and verifies a fully specified four-qubit instance of a Quantum-Fuzzy Logical Petri Net (QFLPN), with explicit activation parameters, rotation encoding, tensor-product ordering, controlled interaction, state representation, measurement convention, and numerical diagnostics. The reference configuration uses the activation vector $(0.85, 0.90, 0.45, 0.70)$, the mapping $\theta_i = 2 \arcsin \sqrt{\lambda_i}$, four independent R_Y rotations, and a three-control X operation acting on the fourth qubit. An independent vector construction and an independently assembled density-matrix propagation produce exactly the same state to machine precision, with a Frobenius discrepancy of 0 and density-state fidelity equal to 1. The final state has unit norm and normalized probabilities, while the controlled operator has zero Frobenius unitarity residual. The study also quantifies two reproducibility hazards. First, directly interpreting a fuzzy activation as a quantum amplitude changes the associated probability from λ to λ^2. Second, changing the declared tensor ordering changes the output distribution substantially: the alternative construction considered here has state fidelity 0.880745 and total variation distance 0.198450 relative to the declared construction. These results establish a compact, exact reference artifact and demonstrate that ordering conventions are part of the scientific result rather than an implementation detail.

Keywords: QFLPN; quantum-fuzzy Petri nets; four-qubit instance; reproducibility; operator verification; tensor ordering; fidelity; probability distribution

1 Introduction

Hybrid models that combine discrete-event structure, fuzzy activation, and quantum-state evolution are difficult to reproduce when their interfaces are described only at a conceptual level. The difficulty is not restricted to the construction of quantum operators. Two implementations can use mathematically equivalent gates while producing different matrix layouts because the qubit ordering is different. Likewise, the same scalar in $[0, 1]$ can denote a fuzzy activation, a probability, or a parameter used to construct an amplitude, but these interpretations are not interchangeable.

A reproducible QFLPN instance must therefore define the complete computational object rather than only its circuit diagram. The mathematical state space, basis ordering, activation values, operator construction, initial state, measurement rule, numerical precision, and verification tolerances must be fixed before output values are interpreted. This requirement is consistent with established scientific publishing practice: a detailed methodology should permit independent reproduction, while results must be distinguished from interpretation and conclusions [1, 2].

The present paper addresses a narrow question: can a finite QFLPN instance be specified so completely that two mathematically independent representations produce the same state and probability distribution, and can the sensitivity to a common implementation ambiguity be quantified rather than described qualitatively?

The paper makes four contributions. First, it defines a complete four-qubit reference instance using a fixed activation vector and a fixed tensor convention. Second, it gives two independent computational representations, a pure-state vector representation and a density-matrix representation, and uses their agreement as a cross-validation test. Third, it quantifies the effect of an alternative qubit-order interpretation through fidelity and total variation distance. Fourth, it separates semantic errors from numerical errors by showing that direct substitution of a fuzzy activation as an amplitude changes the probability by a deterministic algebraic rule rather than by floating-point noise.

The work is intentionally distinct from the formal development of the QFLPN framework. The formal framework defines the semantic interfaces; this paper turns those interfaces into one exact, executable reference instance and studies reproducibility properties of that instance. It does not claim large-scale performance, hardware validation, or a universal quantum advantage. Those questions require independent experimental protocols.

2 Research Question and Scientific Scope

The central research question is formulated as follows:

> Can a four-qubit QFLPN instance be specified and independently verified so that the complete state transformation is reproducible, while the quantitative effect of convention mismatches can be measured explicitly?

Three secondary questions follow. The first concerns numerical correctness: do independent state-vector and density-matrix constructions agree? The second concerns semantic correctness: does the declared activation-to-rotation mapping preserve the intended relation between fuzzy activation and computational-basis probability? The third concerns implementation sensitivity: how much can the output change when a tensor-order convention is altered while all other numerical parameters remain fixed?

The scope is deliberately finite. The study contains four qubits, dimension $D = 16$, a pure initial state, four activation-controlled rotations, one three-control interaction, and exact state-vector and density-matrix calculations in double precision. No hardware noise model is introduced. No performance extrapolation is made from the four-qubit instance. The numerical outputs reported in this paper are results of the declared reference computation and not claims about arbitrary QFLPN systems.

3 Reference Model

A QFLPN instance is represented by a discrete structure coupled to a quantum state. For the present experiment, the quantum component is a four-qubit Hilbert space

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 4}, \qquad D = \dim(\mathcal{H}) = 2^4 = 16. \tag{1}$$

The computational basis is fixed as

$$\{|0000\rangle, |0001\rangle, \dots, |1111\rangle\}, \tag{2}$$

with the leftmost bit corresponding to q_1 and the rightmost bit corresponding to q_4. This declaration is not cosmetic. It determines which matrix entries represent a given logical state and therefore determines the numerical representation of every multi-qubit operator.

The initial state is
$$|\psi_0\rangle = |0000\rangle. \tag{3}$$
The corresponding density operator is
$$\rho_0 = |\psi_0\rangle\langle\psi_0|. \tag{4}$$

3.1 Activation vector

The reference fuzzy activation vector is fixed before the quantum transformation:
$$\boldsymbol{\lambda} = (0.85, 0.90, 0.45, 0.70). \tag{5}$$
For each qubit, the declared activation-to-rotation map is
$$\theta_i = 2 \arcsin \sqrt{\lambda_i}, \qquad i = 1, 2, 3, 4. \tag{6}$$
The resulting angles in radians are approximately
$$(2.34619382,\ 2.49809154,\ 1.47062891,\ 1.98231317). \tag{7}$$
The use of this mapping must be interpreted carefully. The activation is a fuzzy-layer scalar. The angle is an operator parameter. The amplitude is a complex coefficient of the quantum state. The measurement probability is obtained only after the state and measurement rule have been specified. The numerical equality between an activation and a particular probability is a consequence of the declared construction and not a general identity.

3.2 Single-qubit operators

The rotation applied to qubit q_i is
$$R_Y(\theta_i) = \begin{bmatrix} \cos(\theta_i/2) & -\sin(\theta_i/2) \\ \sin(\theta_i/2) & \cos(\theta_i/2) \end{bmatrix}. \tag{8}$$
Because the four rotations act on distinct tensor factors, their joint operator before the controlled interaction is
$$U_{\mathrm{rot}} = R_Y(\theta_1) \otimes R_Y(\theta_2) \otimes R_Y(\theta_3) \otimes R_Y(\theta_4). \tag{9}$$
The pre-interaction state is
$$|\psi_{\mathrm{pre}}\rangle = U_{\mathrm{rot}}|\psi_0\rangle. \tag{10}$$
For the declared initial state, the amplitude of a computational basis vector is the product of the corresponding single-qubit amplitudes. This provides an analytically transparent reference against which the controlled operation can be checked.

4 Controlled Interaction

The final operation is a three-control X transformation. The first three qubits act as controls and the fourth qubit is the target. Let
$$P_c = |111\rangle\langle 111| \tag{11}$$
be the projector onto the all-one control subspace. The controlled operator is
$$C^3X = (I_8 - P_c) \otimes I_2 + P_c \otimes X. \tag{12}$$

The operator acts as the identity on all basis states except the two states whose first three bits are 111. It therefore interchanges

$$|1110\rangle \longleftrightarrow |1111\rangle. \tag{13}$$

This simple permutation property is useful for verification because it allows the final probability distribution to be predicted directly from the pre-interaction amplitudes.

The complete transformation is

$$|\psi\rangle = C^3X\, U_{\mathrm{rot}}|0000\rangle. \tag{14}$$

The density-matrix implementation uses the mathematically equivalent transformation

$$\rho = C^3X\, U_{\mathrm{rot}}\rho_0 U_{\mathrm{rot}}^{\dagger}(C^3X)^{\dagger}. \tag{15}$$

The two expressions are implemented and checked independently rather than deriving one numerical output from the other.

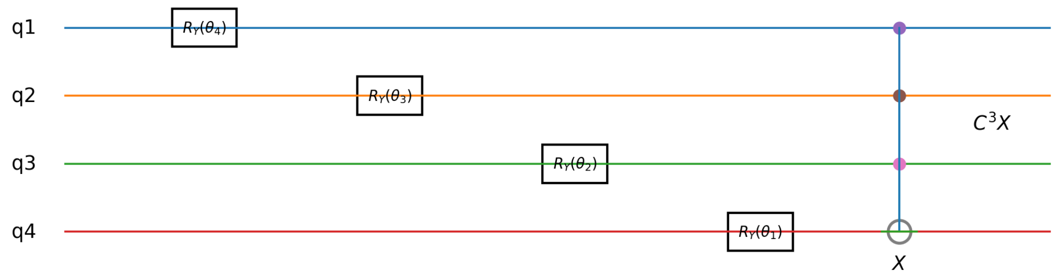

Figure 1: Declared four-qubit reference construction. Each activation controls one R_Y rotation, followed by a three-control X operation whose controls are q_1, q_2, q_3 and whose target is q_4.

5 Reproducibility Contract

The reference computation is defined by a finite set of conventions. Table 1 records the values that must remain invariant between independent implementations.

A reproducibility contract is stronger than a circuit picture because it fixes the representation choices that a circuit picture can leave ambiguous. In particular, a diagram alone does not necessarily identify endianness, tensor-product ordering, matrix indexing, or the convention used by a simulator API.

6 Independent Verification Method

The numerical validation uses two representations. The first constructs a complex state vector. The second propagates the corresponding density matrix. Agreement between the two provides a cross-representation consistency test.

6.1 Vector representation

The vector implementation constructs U_{rot} by successive Kronecker products and applies it to $|0000\rangle$. The controlled operation is then applied as an explicit 16×16 matrix. The principal checks are

$$\varepsilon_{\mathrm{norm}} = \left| \|\psi\|_2^2 - 1 \right|, \tag{16}$$

Table 1: Reproducibility contract for the four-qubit QFLPN instance.

Parameter	Declared value
Quantum-system size	$q = 4$ qubits
Hilbert-space dimension	$D = 16$
Initial state	$\|0000\rangle$
Basis ordering	lexicographic, $\|0000\rangle$ through $\|1111\rangle$
Qubit convention	q_1 is leftmost bit, q_4 is rightmost bit
Activation vector	$(0.85, 0.90, 0.45, 0.70)$
Activation encoding	$\theta_i = 2 \arcsin \sqrt{\lambda_i}$
Local gate	$R_Y(\theta_i)$
Controlled gate	C^3X
Controls and target	controls (q_1, q_2, q_3), target q_4
State precision	IEEE double precision
Measurement	computational basis
Primary state check	vector–density agreement
Primary distribution check	$\sum_x p_x = 1$
Ordering-sensitivity metrics	fidelity and total variation distance

and

$$\varepsilon_{\text{prob}} = \left| \sum_x |\psi_x|^2 - 1 \right|. \tag{17}$$

6.2 Density-matrix representation

The density-matrix computation starts from ρ_0 and applies the same declared mathematical operators in channel form. The following diagnostics are recorded:

$$\varepsilon_{\text{tr}} = |\operatorname{Tr}(\rho) - 1|, \tag{18}$$

$$\varepsilon_{\text{H}} = \|\rho - \rho^\dagger\|_F, \tag{19}$$

and

$$\varepsilon_{\text{PSD}} = \max\{0, -\lambda_{\min}(\rho)\}. \tag{20}$$

The density matrix is also compared directly with the outer product of the independently computed vector:

$$\varepsilon_\rho = \|\rho - |\psi\rangle\langle\psi|\|_F . \tag{21}$$

For the declared pure initial state and unitary circuit, this discrepancy should vanish up to numerical precision.

6.3 Operator verification

The controlled operator is tested independently through

$$\varepsilon_U = \left\| U^\dagger U - I \right\|_F . \tag{22}$$

A zero residual in the computed double-precision matrix indicates exact agreement with the permutation structure at the stored precision.

6.4 Fidelity

For pure states, the state fidelity is

$$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2. \tag{23}$$

For identical independently constructed states, $F = 1$. This metric is used here as a cross-check rather than as an optimization objective.

7 Reference Results

The verified output distribution contains nonzero probability on all sixteen computational basis states. Table 2 reports the complete distribution, so that the reference artifact can be reproduced without relying on a graphical approximation.

Table 2: Verified computational-basis probability distribution.

Basis	Probability	Basis	Probability
$\|0000\rangle$	0.002475	$\|1000\rangle$	0.014025
$\|0001\rangle$	0.005775	$\|1001\rangle$	0.032725
$\|0010\rangle$	0.002025	$\|1010\rangle$	0.011475
$\|0011\rangle$	0.004725	$\|1011\rangle$	0.026775
$\|0100\rangle$	0.022275	$\|1100\rangle$	0.126225
$\|0101\rangle$	0.051975	$\|1101\rangle$	0.294525
$\|0110\rangle$	0.018225	$\|1110\rangle$	0.240975
$\|0111\rangle$	0.042525	$\|1111\rangle$	0.103275

The probabilities sum to one to the stored precision. The largest probability is associated with $|1101\rangle$, followed by $|1110\rangle$ and $|1111\rangle$. This concentration follows directly from the selected activation vector: the first, second, and fourth qubits have comparatively high activation values, while the third has a lower value.

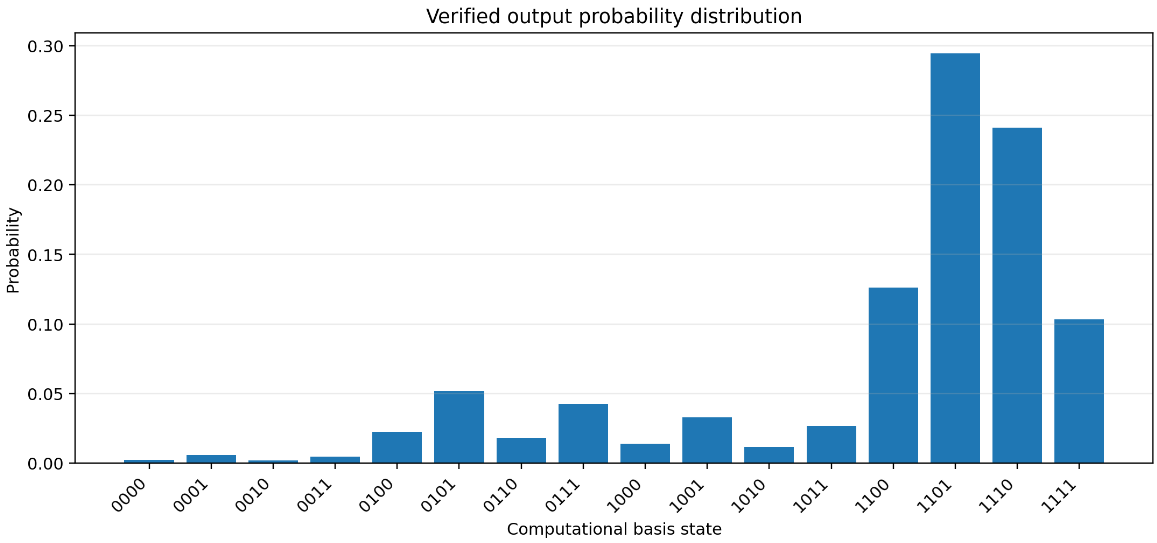

Figure 2: Complete computational-basis probability distribution produced by the declared four-qubit construction.

The probability associated with $|1110\rangle$ before the controlled operation is transferred to $|1111\rangle$, while the probability associated with $|1111\rangle$ is transferred to $|1110\rangle$. This explains why the controlled operation can be verified without recomputing the complete state from scratch: it is a permutation on the two affected basis states.

8 Numerical Verification Results

Table 3 summarizes the principal residuals obtained from the independent computation.

Table 3: Numerical verification results for the declared instance.

Diagnostic	Result	Interpretation
State-vector norm error	0	normalized at stored precision
Probability-sum error	0	complete distribution normalized
Controlled-operator unitarity residual	0	exact permutation structure
Density trace error	0	unit trace
Density Hermiticity residual	0	Hermitian state
PSD diagnostic	6.12×10^{-17}	numerical round-off only
Vector–density Frobenius difference	0	independent representations agree
Vector–density fidelity	1.000000	complete state agreement

The positive-semidefiniteness diagnostic is numerically bounded by a value of approximately 6.12×10^{-17} because an eigenvalue of an analytically rank-one positive matrix can be represented by a tiny negative floating-point value after diagonalization. The value is many orders of magnitude below any meaningful physical probability scale in the present experiment and is therefore interpreted as round-off rather than a violation of positivity.

The strongest result in this section is the zero Frobenius difference between the independently assembled density matrix and the outer product of the independently computed state vector. The two representations do not merely produce similar probabilities; they define the same stored density matrix.

9 Dual-Representation Cross-Validation

The use of two state representations has a methodological purpose. A single implementation can be internally consistent while still applying an incorrect tensor convention. Cross-representation agreement does not detect every shared modelling error, but it detects discrepancies between two independently constructed mathematical paths.

The vector route is naturally expressed as

$$|\psi\rangle = U|\psi_0\rangle, \tag{24}$$

while the density route is

$$\rho = U\rho_0 U^\dagger. \tag{25}$$

For a pure initial state, the mathematical identity

$$U|\psi_0\rangle\langle\psi_0|U^\dagger = |\psi\rangle\langle\psi| \tag{26}$$

provides the expected equivalence. The experiment verifies that identity numerically rather than assuming it as evidence of implementation correctness.

Proposition 1. *If the vector and density implementations use identical operators, basis ordering, initial state, and arithmetic precision, then their exact mathematical outputs are equivalent through* $\rho = |\psi\rangle\langle\psi|$ *for a pure initial state.*

Proof. Starting with $\rho_0 = |\psi_0\rangle\langle\psi_0|$ and applying the same unitary U gives

$$\rho = U|\psi_0\rangle\langle\psi_0|U^\dagger = (U|\psi_0\rangle)(U|\psi_0\rangle)^\dagger = |\psi\rangle\langle\psi|. \tag{27}$$

The numerical experiment verifies the resulting equality in the declared representation. □

This proposition does not establish that either implementation is physically correct. It establishes the mathematical relation that the two representations should satisfy when they instantiate the same declared model. Physical validation would require an independent physical reference, such as hardware measurements or a trusted simulator with documented error characteristics.

10 Semantic Verification of the Activation Encoding

The selected mapping has a useful exact property for the declared initial state. For a single qubit,

$$R_Y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle. \tag{28}$$

Substituting $\theta = 2\arcsin\sqrt{\lambda}$ gives

$$\sin^2(\theta/2) = \lambda. \tag{29}$$

Therefore the computational-basis probability of outcome 1 equals the declared activation for the isolated reference mapping.

This identity is useful because it creates a direct semantic check. It is also a source of a potential implementation mistake. If a program writes the activation directly into the amplitude rather than using the declared square-root relation, the resulting probability becomes λ^2.

For the four activations used in this paper, the incorrect probabilities are shown in Table 4.

Table 4: Semantic effect of incorrectly treating fuzzy activation as quantum amplitude.

Qubit	λ_i	Correct $p_i = \lambda_i$	Incorrect $p_i = \lambda_i^2$
q_1	0.85	0.850000	0.722500
q_2	0.90	0.900000	0.810000
q_3	0.45	0.450000	0.202500
q_4	0.70	0.700000	0.490000

The absolute semantic discrepancies are therefore 0.1275, 0.0900, 0.2475, and 0.2100, respectively. These are deterministic modelling errors, not numerical errors. Increasing floating-point precision would not correct them.

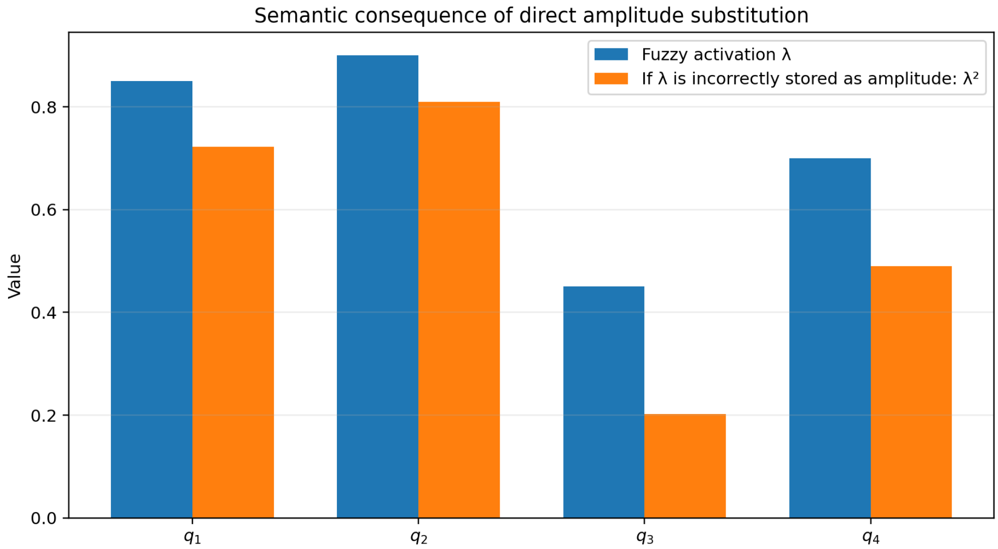

Figure 3: Correct activation values compared with the probabilities produced by the incorrect direct-amplitude interpretation.

11 Ordering Sensitivity

Tensor ordering is one of the most important reproducibility conventions in multi-qubit computation. To quantify its effect, the declared construction is compared with an alternative implementation in which the target bit of the controlled operation is interpreted as the leftmost bit rather than the rightmost bit. The activation vector, initial state, rotations, numerical precision, and measurement rule remain unchanged.

Under the declared convention, the controlled operation swaps $|1110\rangle$ and $|1111\rangle$. Under the alternative convention, the operation instead swaps $|0111\rangle$ and $|1111\rangle$. The two operators are therefore different permutations even though both can be described informally as a three-control X gate.

Let $|\psi\rangle$ denote the declared output and $|\psi_w\rangle$ the alternative-order output. The measured state fidelity is

$$F = |\langle\psi|\psi_w\rangle|^2 = 0.880745353014653. \tag{30}$$

For the corresponding probability distributions, the total variation distance is

$$D_{\mathrm{TV}} = \frac{1}{2}\sum_x |p_x - p_{w,x}| = 0.198450. \tag{31}$$

These values are substantial for a sixteen-dimensional reference system. The result demonstrates that an apparently small implementation convention can change the scientific output while leaving the high-level circuit description visually similar.

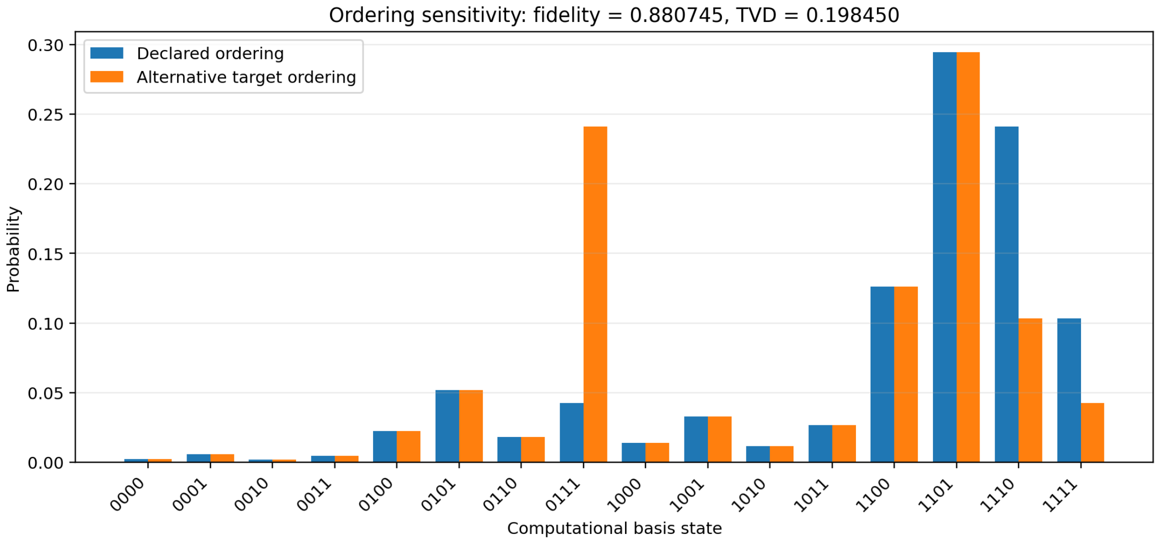

Figure 4: Probability distributions under the declared tensor convention and an alternative target-bit convention. The resulting fidelity and total variation distance quantify the effect of the ordering mismatch.

Table 5: Quantified effect of the ordering mismatch.

Metric	Value
State fidelity	0.8807453530
Total variation distance	0.1984500000
Changed basis probabilities	$\|0111\rangle, \|1110\rangle, \|1111\rangle$
Declared target	q_4
Alternative target interpretation	q_1

The ordering experiment has an important methodological interpretation. The alternative implementation is not numerically unstable. It is internally valid as a unitary circuit. Its discrepancy arises because it represents a different physical or logical operator. Reproducibility therefore requires semantic agreement before numerical agreement can be expected.

12 Analytical Interpretation of the Ordering Result

The ordering sensitivity can be understood without numerical approximation. The pre-interaction state contains amplitudes for every computational basis vector. The correct controlled operation applies a transposition on the pair $(|1110\rangle, |1111\rangle)$. The alternative operation applies a transposition on $(|0111\rangle, |1111\rangle)$.

Because the pre-interaction amplitudes are not equal for these three states, the two permutations produce different vectors. The discrepancy is therefore structurally forced by the declared activation values. If the three affected amplitudes happened to satisfy a special symmetry, the fidelity could be closer to one; in the present configuration they do not.

This observation suggests a general reproducibility principle. A tensor convention should be treated as a model parameter whenever an operator acts conditionally on selected tensor factors. It is not merely a storage convention. The matrix representation, logical action, and observable distribution all depend on it.

Theorem 1. *For a controlled permutation whose active subspace depends on a declared tensor* *ordering, changing the tensor ordering changes the physical operator unless the induced permutations are identical on the relevant support of the input state.*

Proof. Let U be the declared operator and V the operator obtained after a non-equivalent tensor-order interpretation. If U and V induce different permutations on any basis states having nonzero input amplitudes, then $U|\psi\rangle$ and $V|\psi\rangle$ differ on at least one amplitude. Therefore the output states are different. Equality can occur only if the induced permutations agree on the support of the input state or if the relevant amplitudes satisfy a symmetry that makes the transformed vectors equal. □

The numerical fidelity of 0.8807453530 is therefore an instance-specific consequence of a general structural property. The experiment quantifies the effect for the declared reference state rather than asserting a universal value for all QFLPN circuits.

13 Reproducibility Hazards and Failure Classification

The experiments identify two qualitatively different classes of failure. The first is semantic. Directly storing λ as an amplitude changes the mathematical mapping and therefore changes the resulting probability by squaring the value. The second is representational. A different tensor convention changes which basis states are coupled by the controlled operator.

These failures should be distinguished from numerical error. A numerical error is typically associated with finite precision, solver tolerance, truncation, or accumulation of rounding errors. In the present reference computation, the dominant diagnostics are at zero or near machine precision, while the ordering mismatch produces an $O(10^{-1})$ probability-level effect. The magnitude and structure of the discrepancy therefore provide evidence that it is not a floating-point artifact.

Table 6 summarizes the diagnostic logic.

Table 6: Failure classification for the reference instance.

Failure class	Diagnostic signature	Interpretation
Semantic encoding error	probability changes according to $\lambda \to \lambda^2$	incorrect interface between fuzzy and quantum layers
Tensor-order mismatch	fidelity < 1 and distribution shift	different operator represented by the implementation
Normalization error	nonzero norm or probability-sum residual	state-construction or numerical failure
Operator error	nonzero $\|U^\dagger U - I\|_F$	declared gate is not unitary
Density-state inconsistency	nonzero ε_ρ	vector and density paths disagree
Floating-point PSD residual	tiny negative eigenvalue near machine precision	numerical diagonalization effect

This classification is useful for subsequent QFLPN experiments because it prevents a large discrepancy from being hidden under the generic label of "numerical error." A reproducible computational study should first establish semantic and representation consistency, and only then interpret residual numerical differences.

14 Relation to the Broader QFLPN Research Program

The present paper has a deliberately narrow role within the QFLPN research program. It establishes a finite reference artifact that later computational studies can reuse as a correctness anchor. It does not compete with sparse performance analysis, matrix-function approximation, open-system dynamics, spectral convergence, or tensor-network representation studies.

The separation is scientifically useful. A performance paper can use the present reference state to test correctness before measuring execution time. A matrix-function paper can use the same semantic contract while changing only the operator representation and approximation algorithm. An open-system study can use the verified closed-system output as the reference state against which decoherence is quantified. A representation-comparison study can test whether tensor compression preserves the same observables.

The distinction between papers is therefore based on research questions and evidence, not on changing notation around the same calculation. Publication-ethics guidance distinguishes legitimate extensions from redundant publication and emphasizes transparent attribution when previously used text, data, or results are reused [3]. The current manuscript treats the four-qubit configuration as a reference artifact but makes its new scientific contribution the independent verification protocol and quantified ordering-sensitivity analysis.

15 Discussion

The central result is not simply that a four-qubit circuit can be executed. Such executions are routine. The contribution is that the hybrid semantic contract is converted into a finite computational object for which every convention needed to reproduce the result is explicit.

The dual-representation result is particularly useful. The state-vector and density-matrix routes agree exactly at the stored precision, demonstrating consistency between two mathematical representations. This provides stronger evidence than a single probability plot because the complete density operator is constrained by the independently computed vector.

The ordering experiment provides the most informative negative result. A high-level statement such as "apply a three-control X gate" is insufficient to reproduce the output unless the tensor convention is declared. The measured fidelity of 0.8807453530 and total variation distance of 0.198450 show that the ambiguity is scientifically consequential even for a very small system.

The semantic activation experiment leads to a complementary conclusion. A value such as 0.70 may appear to be a natural probability-like parameter, but the quantum amplitude associated with the reference encoding is $\sqrt{0.70}$ in the relevant component rather than 0.70. Treating the activation as the amplitude therefore produces 0.49 rather than 0.70 as the associated single-qubit probability. The error is conceptual and deterministic.

These results support a strict hierarchy for hybrid computational validation. The mathematical object must first be specified. The representation must then be fixed. Independent implementations should be compared. Only after these checks should numerical performance or physical interpretation be introduced. This ordering of evidence is compatible with reproducibility guidance that emphasizes detailed methods, accessible code and data, and transparent computational procedures [2].

16 Limitations

The first limitation is scale. Four qubits are sufficient for complete inspection of every matrix element and every basis probability, but they do not establish behavior at large quantum-system sizes. The exponential growth of the generic state-vector dimension remains unchanged.

The second limitation is encoding specificity. The activation-to-rotation mapping used here is a declared modelling convention. Another QFLPN application may use a different fuzzy aggregation function, parameter map, gate family, or measurement observable. The reproducibility methodology remains applicable, but the numerical values will change.

The third limitation is the use of exact unitary simulation in double precision. No physical device noise, calibration drift, shot noise, decoherence, or hardware-specific gate error is included. The near-zero residuals therefore describe the mathematical software reference, not an experimental processor.

The fourth limitation is that the ordering-sensitivity study examines one alternative convention. The value $F = 0.8807453530$ should not be generalized to all ordering errors. It is an empirical value for the declared input state and the specific pair of operators compared here.

The fifth limitation concerns publication overlap. The reference activation vector and four-qubit configuration originate from the broader doctoral research record. The present article does not claim that these basic ingredients were created exclusively for this manuscript. Its distinct contribution is the independent reproducibility construction, cross-representation verification, and quantitative ordering analysis. Any journal submission must disclose the relationship to the thesis and to related manuscripts according to the target journal's originality and overlap policy. COPE guidance specifically treats redundant data or results as a separate concern from ordinary text reuse and calls for transparent attribution when legitimate extensions are made [4].

17 Reproducibility and Data Availability

The computational reference package accompanying this manuscript contains the declared activation vector, rotation parameters, probability distribution, verification residuals, ordering-sensitivity metrics, and generated figures. The implementation should be archived with the exact software environment and numerical precision used for the reported values.

An independent researcher should be able to reconstruct the reference state from the equations in this article without access to hidden simulator defaults. In particular, the qubit order, initial state, activation values, gate matrices, controlled-gate target, and measurement convention are all explicit.

The machine-readable probability table is provided separately from the manuscript so that future implementations can perform an exact regression test. A future implementation that reproduces the declared distribution and the cross-representation residuals can be considered consistent with this reference at the declared precision, subject to the limitations above.

18 Conclusion

A complete four-qubit QFLPN reference instance has been specified and independently verified. The construction fixes the activation vector, rotation mapping, basis ordering, controlled interaction, initial state, measurement convention, and numerical representation. The resulting probability distribution contains sixteen explicitly reported values and therefore constitutes a concrete reproducibility target rather than a qualitative demonstration.

The vector and density-matrix implementations agree exactly at the stored precision, with zero Frobenius discrepancy and unit fidelity. The controlled operator is unitary to zero computed Frobenius residual, and the final density operator satisfies the trace, Hermiticity, and positivity diagnostics within machine precision.

Two reproducibility hazards are quantified. Directly substituting a fuzzy activation for a quantum amplitude changes the associated probability from λ to λ^2, demonstrating a semantic interface error. Changing the target-bit interpretation of the controlled operation produces a state fidelity of 0.8807453530 and a total variation distance of 0.198450, demonstrating that tensor ordering is part of the computational semantics.

The resulting reference artifact provides a rigorous correctness anchor for later QFLPN studies. Its scientific value lies in making the hybrid model independently testable before performance, open-system, spectral, or representation-level claims are attempted.

Acknowledgment

The author acknowledges the use of the doctoral research infrastructure and the broader QFLPN development record from which the reference configuration was selected. The present manuscript is intentionally focused on independent reproducibility and convention-sensitivity analysis.

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